

PREOBRAZHENKA, Russian Federation

WMO: 249180

Lat: 60.050N

Lon: 108.083E

Elev: 295

StdP: 97.83

Time Zone: 8.00 (E08)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -46.1	(c) -44.0	(d) -48.6	(e) 0.0	(f) -46.1	(g) -46.4	(h) 0.0	(i) -43.9	(j) 6.5	(k) -14.1	(l) 5.5	(m) -18.2	(n) 0.6	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 29.7	(d) 18.4	(e) 27.8	(f) 17.8	(g) 25.6	(h) 16.6	(i) 19.7	(j) 26.8	(k) 18.6	(l) 25.5	(m) 17.6	(n) 24.1	(o) 2.4	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.7	22.4	16.1	11.9	21.4	15.0	11.0	20.2	57.7	26.6	54.2	25.7	50.9	24.2	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.3	(c) 5.4	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 7.4	(5) 6.3	(6) 5.4	(7) -47.8	(8) 22.0	(9) 3.0	(10) 1.4	(11) -49.9	(12) 23.0	(13) -51.7	(14) 23.8	(15) -53.4	(16) 24.5	(17) -55.6	(18) 25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.3	-26.7	-23.7	-11.5	-2.5	5.7	15.4	17.2	14.1	6.4	-4.5	-18.4	-25.0		
	DBStd	17.26	9.45	9.61	8.24	6.53	4.72	4.19	3.15	3.66	4.68	6.06	9.62	10.32		
	HDD10.0	5798	1138	942	668	375	148	6	0	8	127	448	852	1085		
	HDD18.3	8328	1397	1176	926	625	391	107	59	139	357	707	1102	1343		
	CDD10.0	563	0	0	0	0	16	168	225	136	19	0	0	0		
	CDD18.3	51	0	0	0	0	0	18	25	8	0	0	0	0		
	CDH23.3	977	0	0	0	0	7	388	429	146	7	0	0	0		
	CDH26.7	270	0	0	0	0	0	108	124	38	0	0	0	0		
(14) Wind	WSAvg	2.3	2.0	2.1	2.8	3.1	2.8	2.4	2.1	2.0	2.2	2.3	2.4	1.9		
(15) Precipitation	PrecAvg	345	16	11	11	17	36	43	48	53	33	27	25	21		
	PrecMax	463	27	27	34	45	79	80	157	128	63	70	49	39		
	PrecMin	240	8	4	2	2	11	5	5	4	7	8	14	8		
	PrecStd	51	5	6	6	10	19	21	28	24	15	12	8	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.1	-1.9	8.6	14.7	23.9	31.8	32.5	30.3	24.2	10.4	4.5	-1.3		
		MCWB	-5.1	-3.1	2.9	7.2	12.4	18.5	19.1	18.9	15.4	5.2	3.1	-2.2		
	2%	DB	-8.0	-3.5	4.9	11.6	20.3	29.8	29.6	27.2	20.3	7.3	0.5	-5.0		
		MCWB	-8.5	-4.3	1.1	4.9	10.3	17.9	18.9	17.8	13.3	3.9	-0.4	-5.7		
	5%	DB	-10.5	-6.5	2.1	9.2	17.5	27.9	27.8	24.3	17.3	5.1	-2.0	-7.2		
		MCWB	-11.0	-7.4	-0.9	3.5	9.2	17.0	18.3	16.7	11.7	2.7	-3.0	-7.8		
	10%	DB	-12.7	-8.9	-0.4	6.9	14.7	25.6	25.6	22.0	14.3	3.1	-4.4	-9.6		
		MCWB	-13.2	-9.8	-2.6	2.3	7.9	15.8	17.3	15.3	10.0	1.1	-5.3	-10.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.9	-2.8	3.3	7.4	13.7	20.2	21.6	20.3	16.3	6.0	3.6	-1.6		
		MCDB	-4.4	-2.2	6.9	13.7	21.3	27.9	28.2	27.0	22.9	9.1	4.3	-0.7		
	2%	WB	-8.5	-4.3	1.2	5.3	11.3	18.9	20.1	18.7	14.2	4.4	-0.4	-5.4		
		MCDB	-8.0	-3.2	4.5	10.3	18.2	27.7	27.4	24.7	19.0	6.9	0.4	-5.0		
	5%	WB	-11.0	-7.4	-0.6	3.9	9.7	18.0	19.1	17.4	12.3	2.9	-3.0	-7.8		
		MCDB	-10.5	-6.5	1.9	8.6	16.8	25.9	25.4	22.7	16.1	5.1	-2.0	-7.3		
	10%	WB	-13.2	-9.7	-2.4	2.6	8.4	16.9	18.0	16.3	10.4	1.4	-5.3	-10.0		
		MCDB	-12.7	-8.9	-0.2	6.5	14.1	24.0	24.4	21.0	13.4	2.8	-4.4	-9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	13.4	14.5	12.4	11.3	13.8	12.8	12.4	10.6	7.5	9.3	9.0		
		MCDBR	11.0	9.6	13.5	13.7	15.9	18.5	17.4	17.0	16.3	9.8	8.8	8.3		
	5% WB	MCWBR	10.8	8.8	10.2	8.7	8.3	8.2	8.1	8.5	9.4	6.6	8.5	8.1		
		MCWBR	10.9	9.5	12.9	12.1	14.2	16.2	15.1	14.7	14.1	8.8	8.7	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.289	0.312	0.343	0.372	0.402	0.461	0.425	0.339	0.305	0.251	0.225		
	taud		1.962	2.010	2.050	2.112	2.295	2.314	2.132	2.197	2.435	2.301	2.009	1.905		
	Ebn at Noon		481	680	799	847	853	829	759	758	787	680	477	353		
	Edh at Noon		49	85	112	128	116	116	137	118	79	64	46	34		
(44) All-Sky Solar Radiation	RadAvg		0.50	1.46	3.05	4.72	5.19	5.78	5.21	4.10	2.52	1.17	0.59	0.28		
	RadStd		0.03	0.09	0.18	0.29	0.36	0.43	0.46	0.33	0.22	0.14	0.06	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+2.29	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page